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CS 520 - Final Project Plan

Topic: MSR 2027 Mining Challenge

Dataset: GitSkills

Research Questions:

- Do the linguistic properties of newly written skills change across monthly cohorts, in structure and phrasing as well as in topic coverage and semantic diversity? Convergence toward formulaic templates would indicate an emerging genre; shrinking diversity may also reflect rising machine authorship.
- How many skills do agents themselves create or maintain, and in which natural languages are skills written? A skill is read by a multilingual model, so a developer may state a procedure more precisely in their own language than in English.

Experiments:

- **E0 - Data preparation**: Parse every SKILL.md into frontmatter, body, headings, and code blocks. Choose the unit for analysis, and assign each skill a cohort month by earliest appearance, and flag catalog or template repos.
- **E1 - Natural-language distribution**: Detect the language of each skill's description, body prose, and paragraphs. Report language shares both overall and by cohort, and check whether descriptions and bodies use the same language.
- **E2 - Agent authorship and maintenance**: Estimate how many skills agents create or maintain by combining markers. Compare first and last commit authors to separate creation from maintenance, and report range.
- **E3 - Structural convergence**: For each cohort, measure size, heading counts, recurring sections, templates, etc. to see whether skills are converging on shared structures.
- **E4 - Phrasing convergence**: Track lexical diversity, rising and falling n-grams, imperative/second-person style, boilerplate, and the description field.
- **E5 - Topic coverage and semantic diversity**: Embed and cluster distinct skills, compare the cohorts on pairwise distance, cluster entropy, and how many clusters cover 80% of skills, with equal sized samples. Track topic growth as well.
- **E6 - Integration**: Re-run E3-E5 split by authorship class and language, fit regressions using cohort. This will test if convergence persists among human-like skills or only in agent-like ones.

Anticipated Challenges:

- Not every skill has a clear "first written" date and commit histories may be samples. Check what dates exist first, and then run the analysis on the subset with good dates.
- Half the files are verbatim copies, so popular templates make skills seem more similar than they are. Analyze distinct contents as the main unit and report the copy-inclusive view separately.
- One bulk-generated or catalog repo can add thousands of near-identical skills in a single month. Cap each repo's contribution and rerun with and without them.
- Required frontmatter makes skills look alike by design. Measure convergence in the optional parts and treat frontmatter as a baseline.
- A classifier might just learn "recent" or "popular". Use explicit markers as a floor, validate against hand-labeled skills, control for cohort, report a range.
- Many agent-written skills are committed under a human's name, so explicit markers may undercount. Use the classifier as the upper bound.